Novel MM-Wave EPR Spectrometer ELEXSYS 7th Series: Design and Performance

Gromov I, Maixner M, Prisecaru I, Bauer K, Schulte J, Tavernier A, Schmalbein D, Höfer P.

Bruker Biospin GmbH, Rheinstetten, Germany

Vonlanthen P, Hinderer J, Bär A, Mock P, Eckert D, Schett O, Jeker R.

Bruker Biospin AG, Fällanden, Switzerland

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1 Introduction

Over the last two decades high frequency electron-paramagnetic resonance (EPR) spectroscopy in sub-terahertz range (>100 GHz) has made considerable progress [1, 2, 3]. The principal enhancement in spectral and orientation resolution, and motional dynamics in conjunction with the explosive interest in dynamic-nuclear polarization technology in high resolution NMR [4] are the driving forces of this development. An increasing number of ESR spectrometers were developed employing special microwave sources, resonators, and quasi-optical techniques [5] working in continuous-wave (CW) [6, 7, 8, 9] and pulse mode [10, 11, 12, 13]. In particular, low loss corrugated transmission lines, polarization based diplexers, and high-frequency multipliers in combination with low phase noise sources are their key features. This combination was found to be well suited and reliable for CW and pulse EPR spectroscopy.

2 Design

The E700 (CW) and E780 (CW/FT) spectrometer utilizes recent achievements in the millimeter wave technique in combination with the Bruker Intermediate Frequency (IF) concept. The system layout is shown in Figure 1. The IF bridge mounted on left side of the system frame operates at X-band frequencies. The quasi-optical front-end rolling on top of the frame converts the IF bridge frequencies up to 263 GHz for excitation and it down converts the resulting signals to X-band frequencies for quadrature detection. As a member of the ELEXSYS line of spectrometers, the E700/E780 spectrometer is controlled via the Xepr software package.

The E700/E780 features two detection modes for CW and pulse operation: reflection and induction. In reflection mode, the reflected signal (co-polarized to excitation wave) from the

sample is doubly down-converted to DC using the internal reference arm. In induction mode, the induced signal (cross-polarized to excitation wave) from the sample is down-converted to DC either using the internal reference arm or the reflected signal from the sample. The latter method allows for artifact compensation due to propagation delay variation over the transmission line. The co- and cross-polarized waves are decoupled which makes induction mode particularly useful in CW operation through a reduction of source noise contribution in the signal. In both detection modes, the signal is detected in quadrature providing the real/imaginary, dispersion/absorption in CW, signals.

The E700/E780 incorporates two probeheads different in sample head – a wide sample space (WSS) and a single mode TE_{011} resonator. Both probeheads use the similar base consisting of thin walled nickel silver corrugated feed line 18 mm in diameter and taper to 5 mm. The probe-heads have two mechanical tunings, three times two electrical lines, two RF feeds and optionally up to two optical fibers. The corrugated line is machined directly in 16 cm sections which are laser welded. The probeheads have also a common head design accommodating gas flushing options and a sealed window. The overall length of the base corrugated transmission line is about 80 cm and total losses are ≤ 0.1 dB.

For ease of handling the E700/E780 system features two front-end/probe head positions. In one position, the probe is mounted outside the magnet and the quasi-optical front-end is positioned above the probe head. In this position, sample mounting and fine tuning can be accomplished. For EPR measurements, the probe head is mounted in the sample cryostat in the magnet and the quasi-optical font-end is positioned above the probe head as shown in Figure 1. The front-end is movable in the horizontal plane and can be tilted in two orthogonal vertical planes. The frame is mechanically decoupled from the magnet system eliminating possible vibrations induced by the cryocooler.

2.1 Quasi-optical front-end

The quasi-optical front-end contains the electronics for converting the IF frequencies to 263 GHz and the necessary mm-wave quasi-optical components for directing the mm-waves for excitation and detection (Figure 2). The front-end receiver is built on a heterodyne scheme where two microwave frequencies are synthesized simultaneously – RF and reference signals. The synthesis is based on a digital approach where products of dividing the input frequency are combined with the local oscillator frequency and then multiplied by eight in the signal pathway on the multiplier chain (>30 mW at 263 GHz) and by four in the reference arm. SH1 and SH2 are sub

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harmonically pumped mixers and consequently the overall multiplication in the reference channel is eight as well. The following formula describes the applied conversions: $f_{RF} = 8 \times f_{LO} + 5 \times f_{IF}$, $f_{REF} = 8 \times f_{LO} + 4 \times f_{IF}$. The mixers have 9 dB SSB conversion losses and 2-12 GHz output bandwidth. The local oscillator LO has a phase noise of -117 dBc and the phase noise for f_{IF} is about -120 dBc. The long term stability of the LO's frequency 26.88 GHz is ±1.5 ppm over eight hours.

The RF signal radiated by the multiplier (red arrows in Figure 2) is directed via a corrugated horn to the probehead (perpendicular to picture plane) by a pair of elliptical off-axis mirror which provide optimal coupling of the Gaussian beam to the hybrid mode HE_{11} in corrugated waveguide. Two wire-grid polarizers P2 and P3 in combination with the Faraday plate FAR form a quasi-optical circulator. A remotely controlled free standing wire-grid P1 regulates the power in the excitation path in the range 0 to <-25 dB. Polarizers P2 and P3 are set to pass the excitation signal. Consequently a cross-polarized signal induced by EPR (blue arrows) is reflected from P3 and routed to the SH1 mixer. A co-polarized beam coming from the probehead (green arrows) experiences another polarization shift in the Faraday plate and after reflection from P2 arrives at the SH2 mixer.

The quasi-optical element and their mutual positions were simulated and optimized to have good isolation between co- and cross-polarized beam components as well as low standing waves. Experimentally the isolation was found to be > 25 dB depending on which type of load was used– flat or rooftop mirror, and corrugated waveguide shortened at the end. All polarizers used have a 50 mm aperture. To measure the overall losses over the signal pass flat or roof-top mirrors were placed in focal plane of mirror M2 (equal to the upper plane of the corrugated waveguide). Total losses from the source to SH1 (induction channel) and to SH2 were about 5 and 6 dB, respectively.

2.2 IF bridge 263 GHz

The IF bridge has two microwave synthesizers tunable in the range 9.65±0.4 GHz. The first synthesizer works as a source for the main microwave channel and the second one is used for two frequency experiments as double electron-electron resonance. The main channel includes a voltage controlled oscillator (VCO) which provides a fast frequency ramp to monitor signals during probehead adjustment. All input signals are directed to the low noise converter (LNC) combining amplifier, double-balanced quadrature mixer and coupler ended with a diode detector.

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The latter provides a smooth phase independent tune picture. Three diode switches select between reflection or induction measurements using internal reference, or induction measurement utilizing the reflected signal as a reference for the LNC. In both cases the level and the phase of the reference signal can interactively be adjusted. The internal frequency counter delivers instant microwave frequency reading and is used to match the VCO center frequency to the synthesizer frequency. Two high speed switches make available pulse operation in the main and auxiliary microwave channels. The receiver path works in the same way for pulse and CW operation. Entire signals DC, pulse and CW (6-100 kHz modulated) are provided by the bridge for further processing by dedicated devices.

2.3 Wide sample space probehead

The E700/E780 wide sample space probe head is based on a corrugated transmission line and works in a non-resonant and resonant configuration (Figure 3). The cylindrical sample area is up to 3 mm in length and 5 mm in diameter. In the simplest configuration liquid, frozen or powder samples in a standard X-band sample tube are placed from the bottom of the probehead. Alternatively, the samples can be placed in a Teflon container which is then mounted on a Rexolite sample holder for vertical positioning of the sample. The sample holder ends with a flat cooper-beryllium mirror and is screwed to the moving frame. Finally a mesh grid can be inserted at the end of the feeder line which in combination with a flat mirror in the sample holder forms an oversized resonator. The latter configuration allows for improved resonator sensitivity. The differential drive insures smooth frequency tuning in resonance configuration and allows adjusting the sample position in the non-resonant mode. The sample insert is equipped with modulation coils which can provide up to 50 G peak-to-peak magnetic field at 100 kHz.

2.4 Cylindrical resonator

The TE_{011} ENDOR probe head has a corrugated transmission line ending with a taper to rectangular waveguide WR3. The TE_{011} resonator (Figure 4, courtesy Helmholtz Zentrum Berlin) has variable frequency tuning and coupling accessible via high ratio warm gear mechanisms. The probe incorporates both ENDOR and CW modulation coils and, optionally, optical access for sample illumination.

The resonator is formed by a cylinder of one mm in diameter and two plungers which go into the cylinder. The plungers are placed at the factory to provide proper frequency and coupling range. Fine frequency tuning is done by moving a quartz capillary tube along the cavity axis (on back

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side in Figure 4). The sample tube 0.33/0.2 mm outer/inner diameter (the sample volume is0.03 mm^3) is inserted from the front side and goes into a tuning capillary. Also that capillary provides smooth and fine frequency adjustment, and increases the magnetic field at the sample, and reduces the influence of the sample on the resonator frequency. The coupling is done via a smoothed rectangular window in the side wall of the cylindrical resonator. It is tunable by shifting the cavity housing with respect to the waveguide. Inset in Figure 4 shows typical resonance curve of TE_{011} cavity.

2.5 Cryofree superconductive magnet

The EPR 12T CF magnet is a completely cryogen liquid free system. It has a closed cycle cooling refrigerator consisting of a Gifford-McMahon cryocooler, and a water cooled compressor. The refrigerator maintains the magnet temperature at ≤ 3 K. The magnet has a main coil and a sweep coil. The main coil produces a homogeneous field of up to 12 T. The field can be set with 21-bit resolution. The main coil has a persistent switch which allows a very high stability of the magnetic field. The sweep coil generates a homogeneous magnetic field of ~ ± 0.15 T on top of the background magnetic field of the main coil with an extremely high resolution of 19-bit. The sweep coil is permanently connected to power supply. Other specifications of the magnet are listed in Table 1.

Description	Value	Unit
Homogeneity in 10 mm DSV(diameter of spherical volume)	≤ 10	ppm
Energizing time to 12 T	< 100	min
Field stability in persistent mode	≤ 2	ppm/day
Radial fringe field (horizontal distance of the 0.5 mT line from the magnetic center)	< 3.4	m
Axial fringe field (vertical distance of the 0.5 mT line from the magnetic center)	< 4.3	m
Magnet free bore	80	mm

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Main coil magnetic field/ current	0.1	T/A
Main coil voltage rate	3.2	V/min
Sweep coil magnetic field/current	3.6	mT/A
Sweep coil maximum current	40	A
Sweep coil maximum rate	40	A/min
Cool down time from room temperature	<96	h

To improve the linearity of magnetic field sweeps with the super conducting magnet, the E700/E780 spectrometer features a calibrated field sweep mode. This mode provides a high precision field sweep for reliable signal averaging and hyperfine measurements. The magnetic field settings are corrected during field sweep using a calibration table for field hysteresis in up and down directions reducing field nonlinearity by one order of magnitude to 10 ppm.

3 Representative Results

In Figure 5 absorption (bottom) and dispersion (top) CW spectra of Mn^{2+} in CaO powder at room temperature are shown demonstrating a perfect baseline and phase stability. The next example (Figure 6) shows the spectrum of phosphorus doped silicon (sample courtesy M. Fanciulli) measured at 10 K were it becomes dielectric. In both cases the WSS probehead was used in the non-resonant configuration and samples were placed in X-band sample tubes.

To characterize the spectrometer sensitivity in CW mode a low concentrated solution of TEMPOL, 111 µM (20% glycerol in water) was used. In Figure 7 two spectra are shown measured at 50 K using the WSS probe in resonant (top) and non-resonant (bottom) configurations. In both cases a short conversion time of 80 milliseconds was used. The observed signal-to-noise ratio is about 120 in the resonant configuration and the minimal sensitivity is found to be 10 nM/mT/$Hz^{1/2}$. The latter number demonstrates that the spectrometer has better or similar CW sensitivity compared with that reported for similar setups [9, 12]. Pulsed experiments are possible with the WSS probe as well. The $\pi/2$ pulse is about 400 ns long as one can see from electron spin nutation experiment (Figure 8, green trace). The optimal pulse length required by the WSS probehead is relatively long limiting the excitation bandwidth and consequently the

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signal amplitude, however, the available sample active volume of 25 mm^3 compensates the signal loss considerably. A case in point is the pulse EPR of the dangling-bond defects in silicon shown in Figure 9 (data courtesy Helmholtz Zentrum Berlin). In addition to a clear spectrum shown in the inset, the spin echo recovery time was measured up to 1 second time scale demonstrating perfect system stability.

The typical π/2 pulse length using the TE_{011} ENDOR resonator was found to be 50 ns (Figure 8, red trace). Taking into account the 30 mW power, 3 dB losses in the quasi-optical circuits and a quality factor Q of 500, we obtain a cavity conversion factor $\frac{B_1}{\sqrt{QP}} \approx 7.2\ G/\sqrt{W}$. This performance is similar to that reported earlier for the pulse TE_{011} resonator at 275 GHz [10] and better than that shown by the ENDOR version [11]. This high conversion factor allows one to easily record pulse field swept spectra at low sample concentrations as demonstrated with the pulse EPR spectrum of 33 μM tyrosyl radical in Y122-RNR frozen solution at 10 K (insert in Figure 10, data courtesy MPI Göttingen). The sample measured with a 500 μM concentration shows an amazing signal-to-noise ratio even in the first-derivative representation of the field swept pulse EPR spectrum (Figure 10, green and red line correspondingly). Comparison of the TE_{011} and WSS resonator (in TEM_{006} mode) shows that the first provides about an order of magnitude higher microwave field at the sample whereas the latter one has a thousand times higher volume. Therefore the first one has a high absolute sensitivity and the second one ahigher concentration sensitivity.

An important issue at millimeter waves is the stability of the system. In particular, frequency and phase deviation, as well as mechanical oscillation leading to propagation delay variation and consequently phase shifts manifesting as dispersion contamination in absorption spectra and vice versa. The stability was checked by recording the line shape variation of a single line of $Mn2^+$ in CaO of 0.8 G line-width over time. The observed spectral variations expressed in terms of phase shift were found to be < 15° over one hour.

Finally pulse ENDOR is demonstrated over a wide frequency range with deuterium in BDPA polycrystalline powder at 61 MHz, lithium at 155 MHz and fluorine in LiF powder at 376 MHz showing perfect quality with a 125 W amplifier and 40 μs pulse length (not optimized) (Figure 11, left-to-right). The shortest π pulse obtained for protons is 25 us.

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4 Conclusion

The E700/E780 spectrometer covers the majority of demands in EPR at millimeter waves while demonstrating excellent performance in both CW and pulse experiments. Due to the cryofree design, the system does not required regular maintenance and allows for sweeping of the main magnetic field without extra helium consumption. The system philosophy permits application specialists to operate the instrument easily and opens mm-wave area to the widest group of EPR users. Further developments include a 10 W power amplifier at 263 GHz which allows one to combine the large sample space and the short microwave pulses to increase the spectrometer capability above any present in this frequency range.

Acknowledgments

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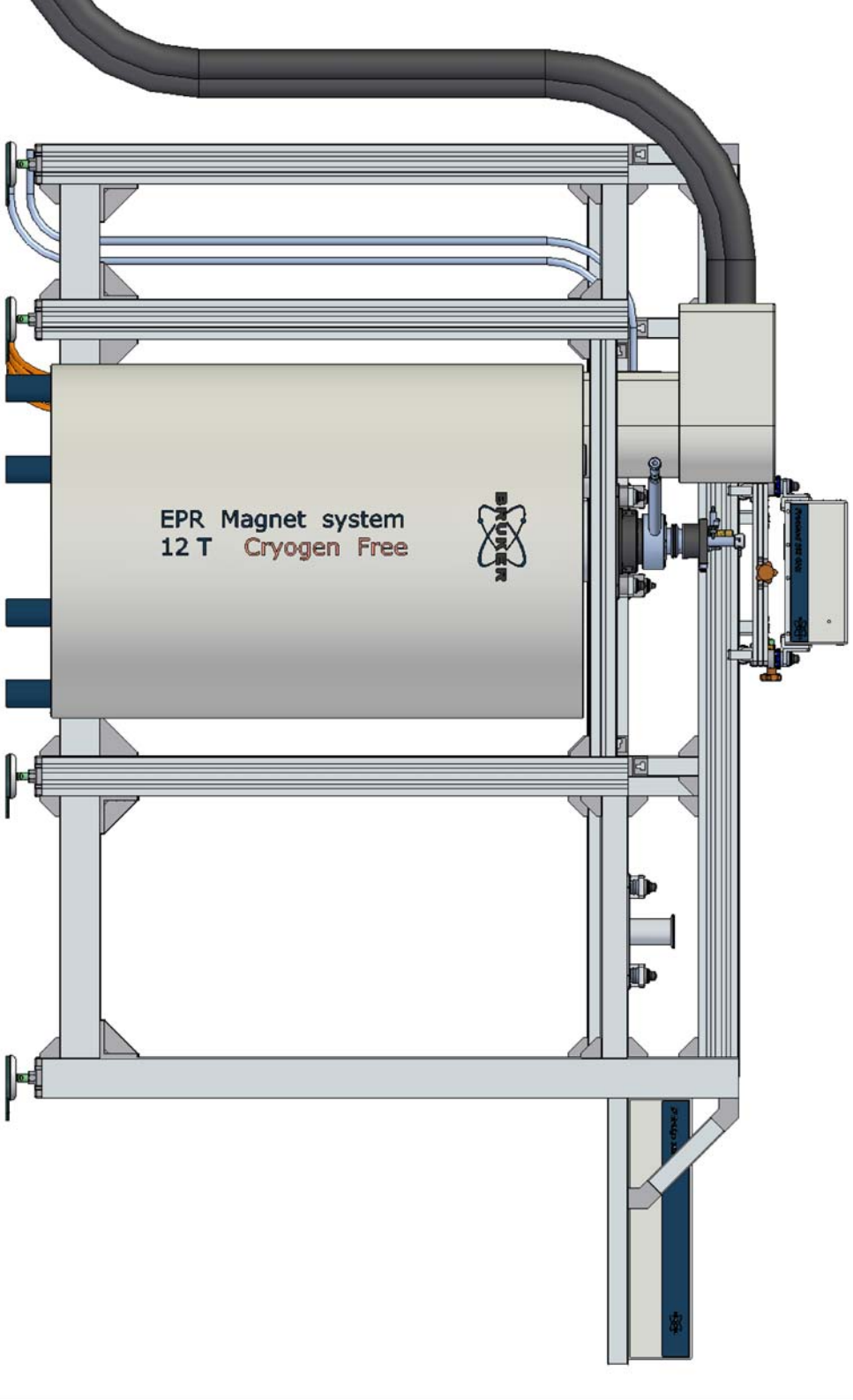
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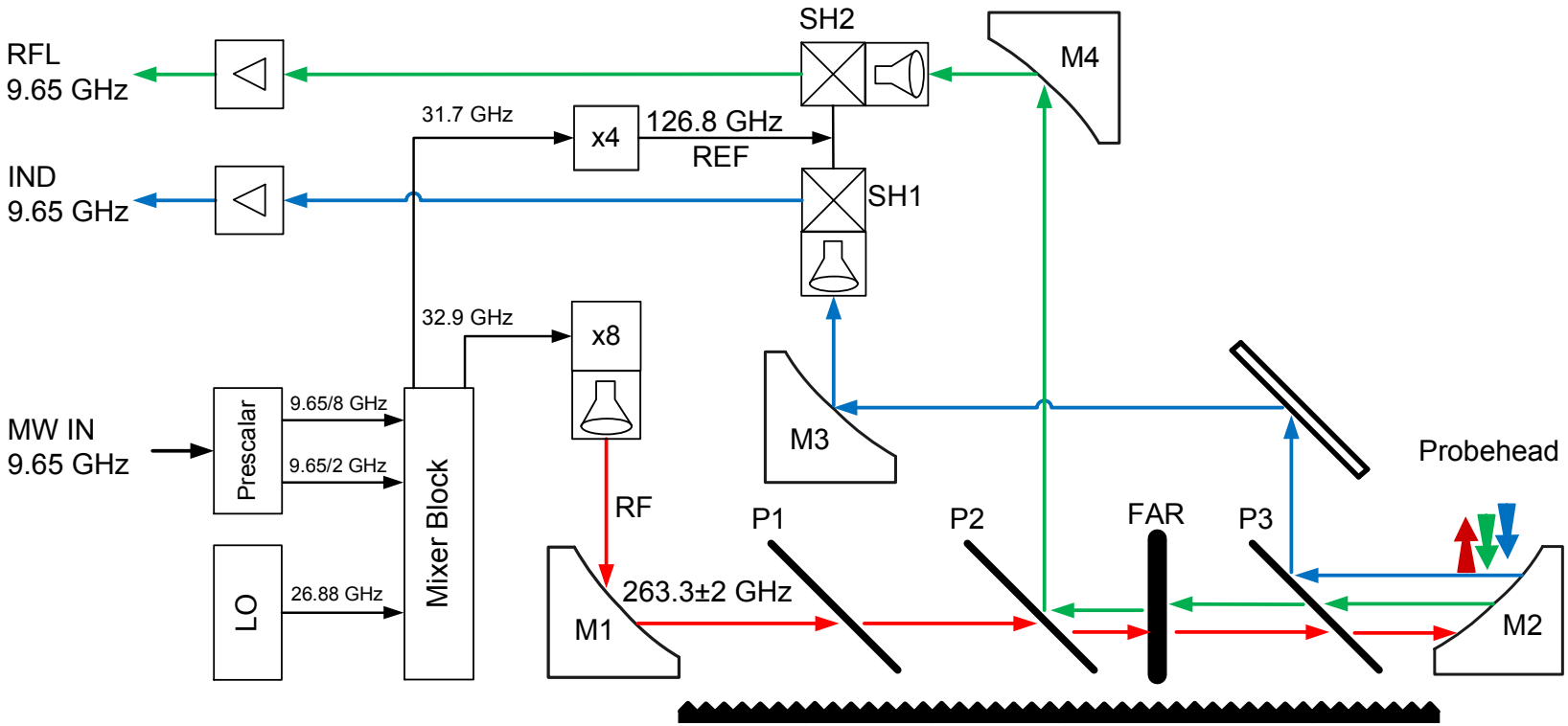
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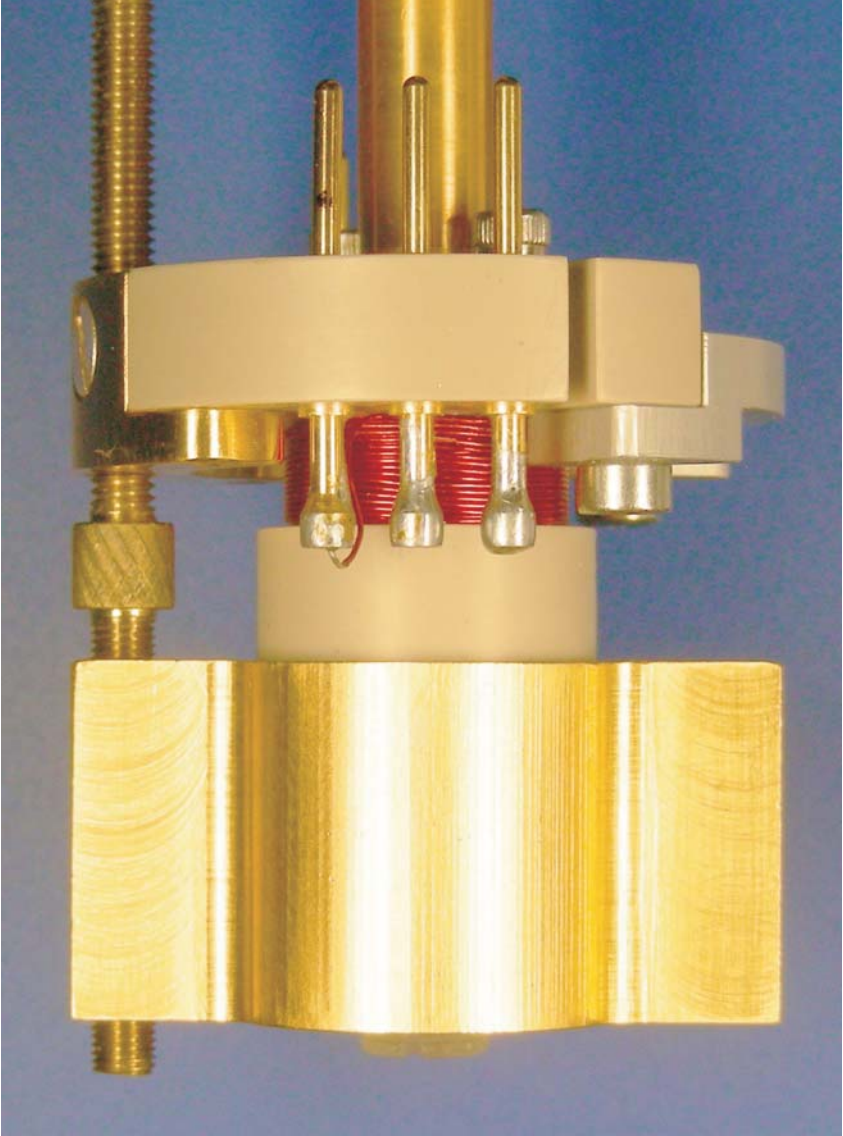
Figure captions

1. E700/E780 system consisting of cryogenfree 12 T magnet with cryocooler, sample cryostat and probehead, quasi-optical mm-wave frontend and microwave IF bridge.
2. Block diagram of quasi-optical frontend showing up/down signal conversion and beam propagation. The excitation shown in red, reflected wave in green, and cross-polarized (induced by EPR) signal in blue.
3. Wide sample space cavity connected on the top to corrugated line. Sample insertion is made from the bottom.
4. Single mode TE_{011} ENDOR resonator (courtesy Helmholtz Zentrum Berlin). Inset shows resonance curve on spectrometer Tune monitor (horizontal span is about 4 GHz).
5. CW spectra of Mn^{2+} in CaO powder recorded in absorption (bottom) and dispersion (top) mode at room temperature using WSS probehead.
6. Phosphorus donors in silicon measured at 10 K in CW mode with WSS probehead.
7. CW spectra of frozen solution 111 μM TEMPOL water/glycerol 20% obtained with WSS probehead in resonant (upper) and non-resonant (bottom) configurations at 50 K.
8. Electron spin nutation acquired with TE_{011} cavity (red) and WSS resonator (green)
9. Recovery curve of the electron spin of dandling bond defects in silicon after inversion pulse at 5 K. Inset shows two pulse electron-spin echo field swept spectrum. The WSS probehead is in use. Data courtesy Helmholtz Zentrum Berlin.
10. Two pulse electron-spin echo field swept spectra of tyrosine radicals in Y122-RNR (upper) and its first derivative (bottom) at 10 K. TE_{011} resonator in use. Data courtesy MPI Göttingen.
11. ENDOR spectra of deuterium in BDPA (left), ^{7}Li in LiF (middle) and ^{19}F in LiF(right).

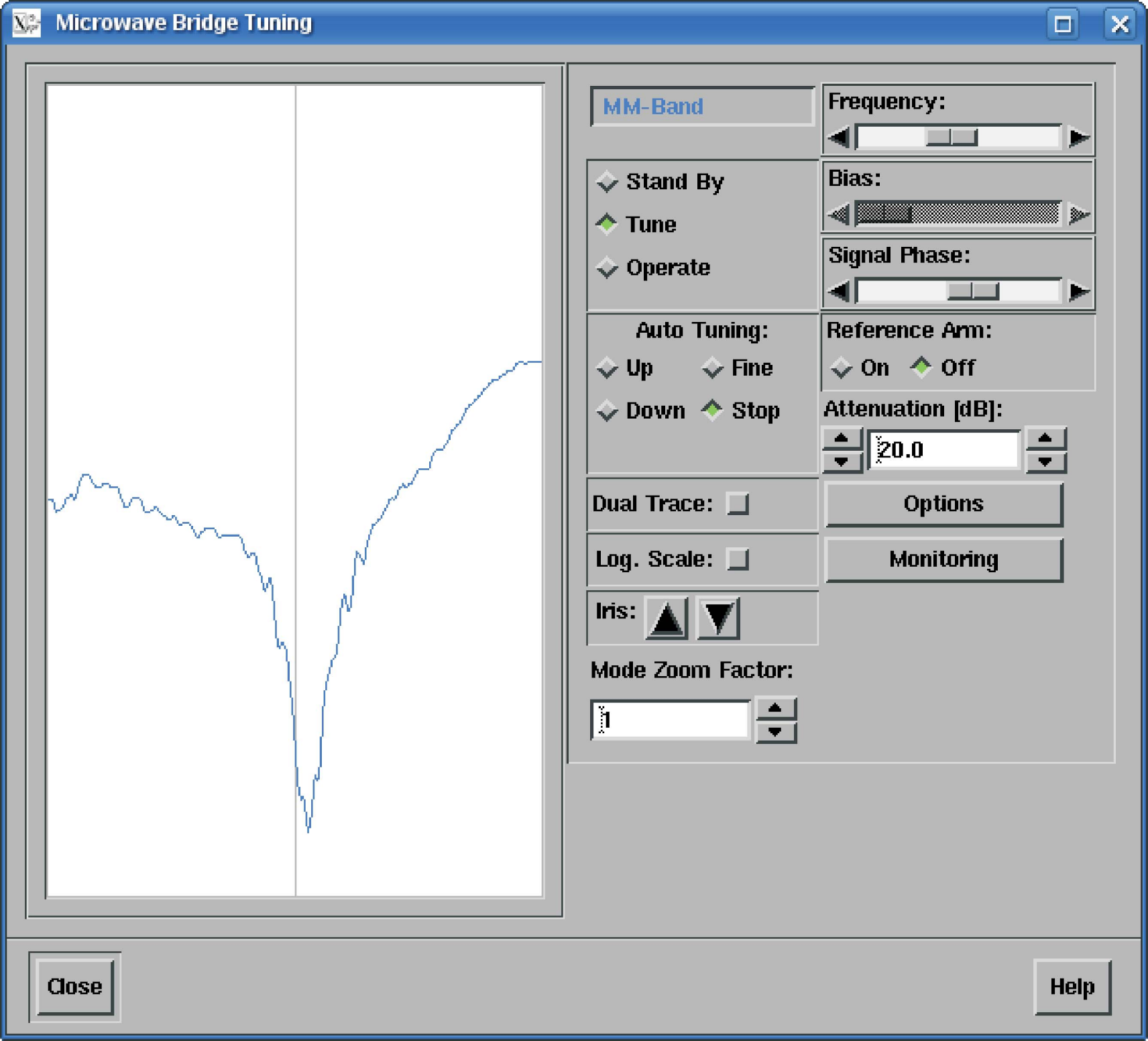
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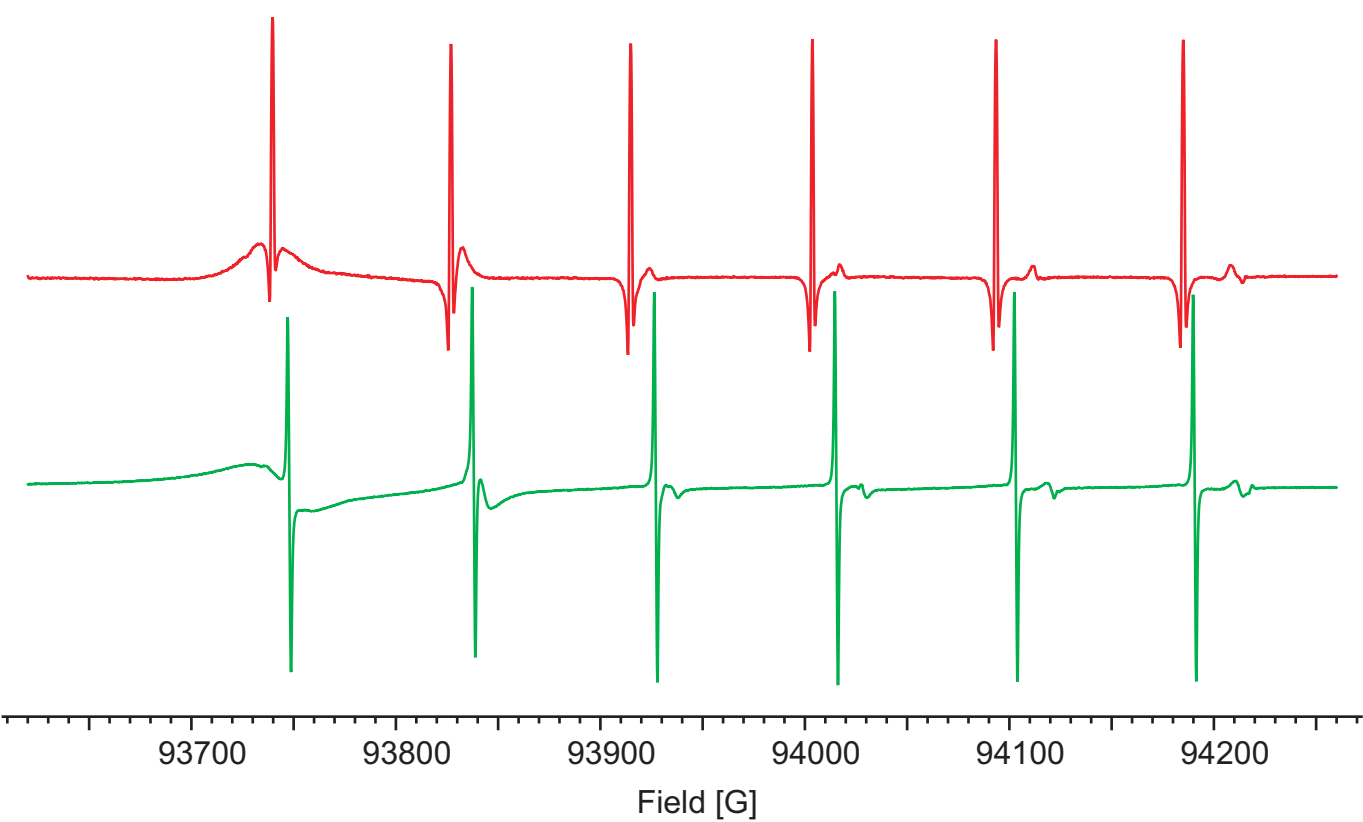

EPR Magnet system
12 T Cryogen Free
BRUKER

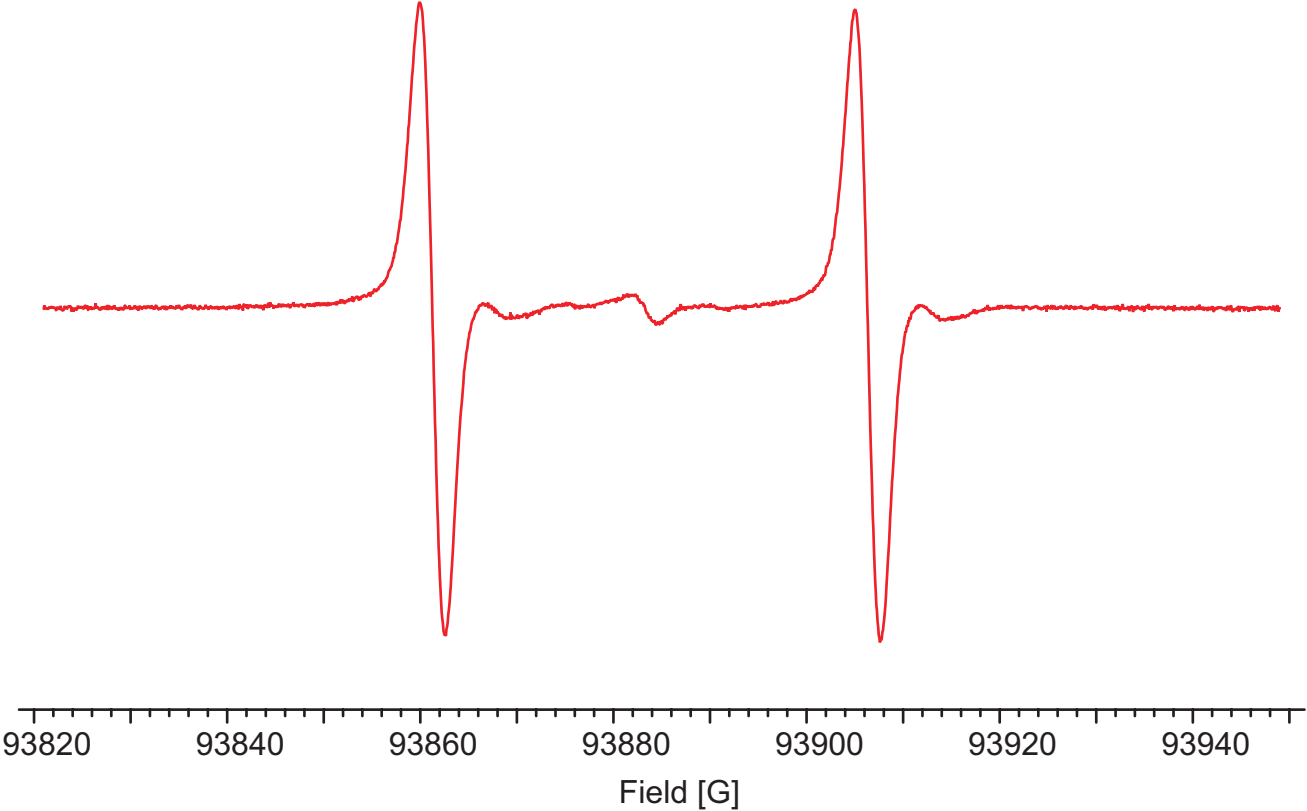

RFL
9.65 GHz
IND
9.65 GHz
MW IN
9.65 GHz
Prescalar
9.65/8 GHz
9.65/2 GHz
LO
26.88 GHz
Mixer Block
31.7 GHz
x4
126.8 GHz
REF
32.9 GHz
x8
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Probehead

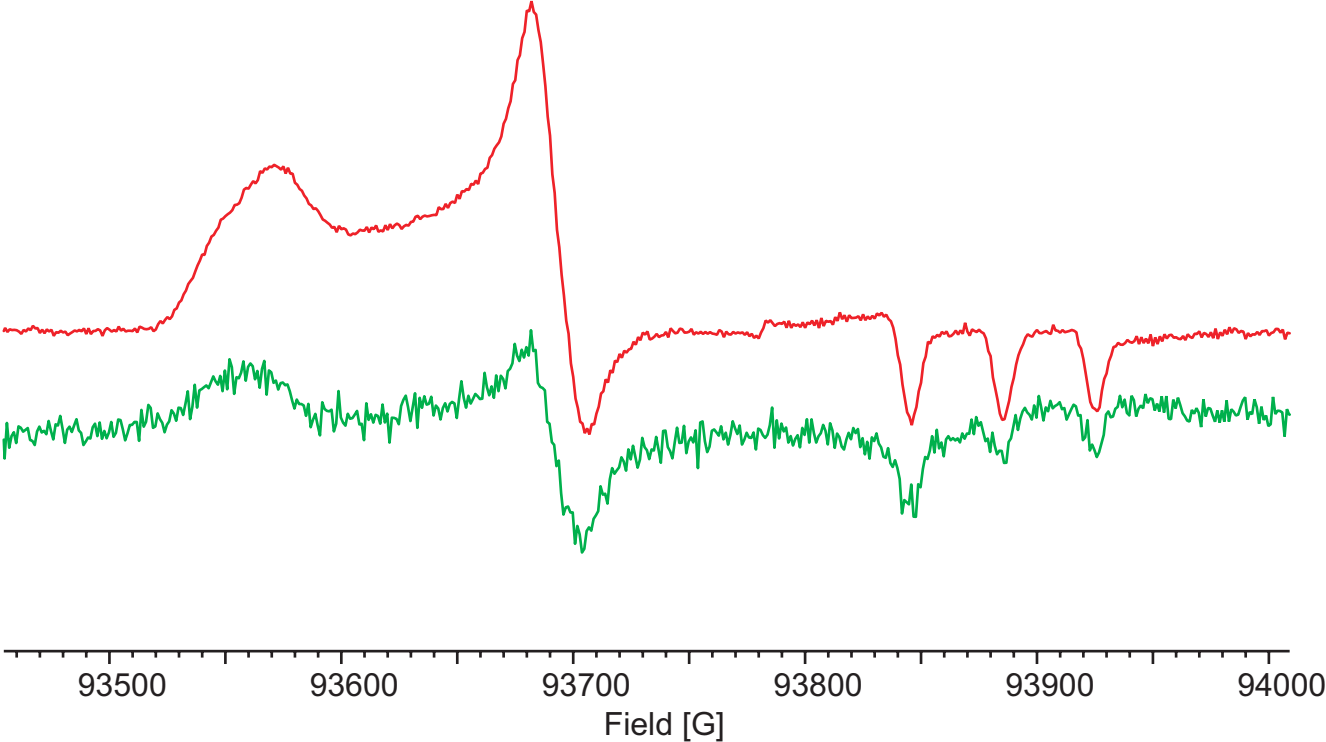


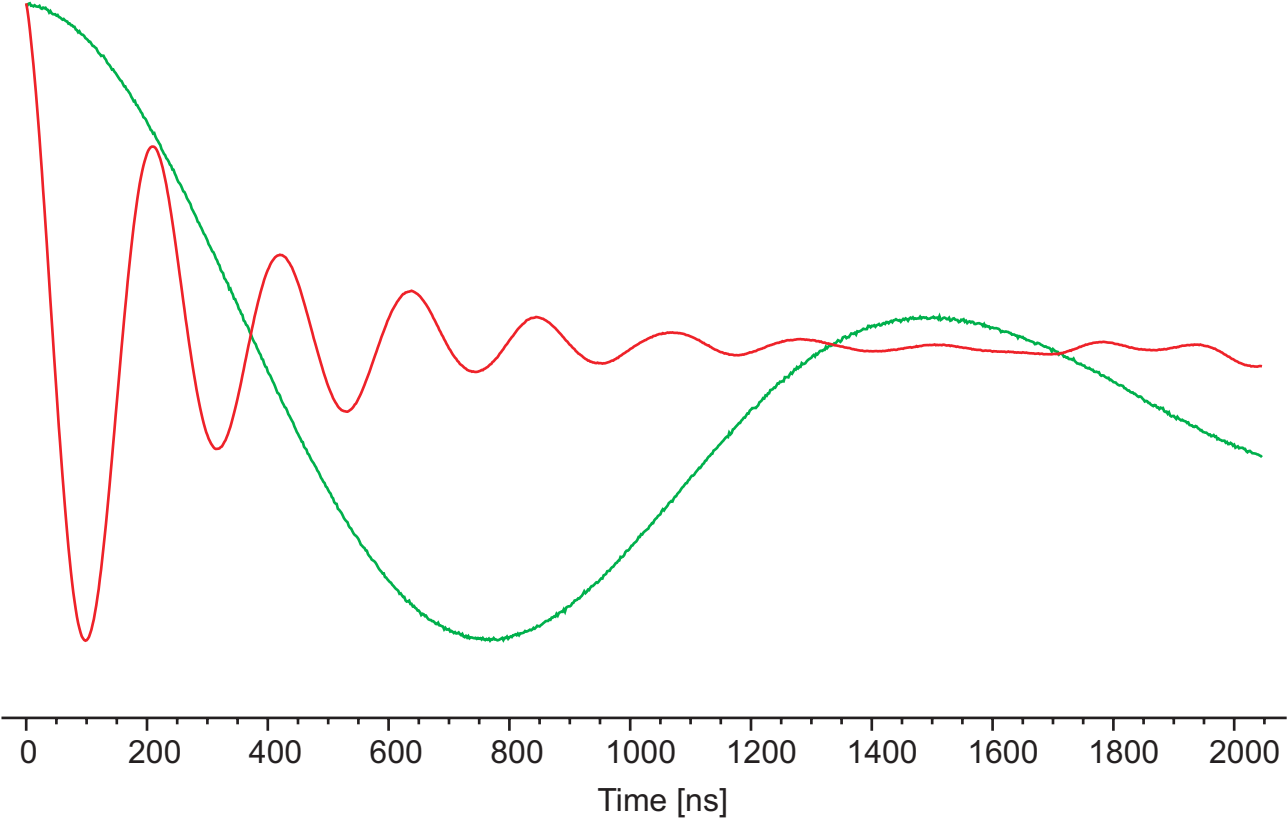

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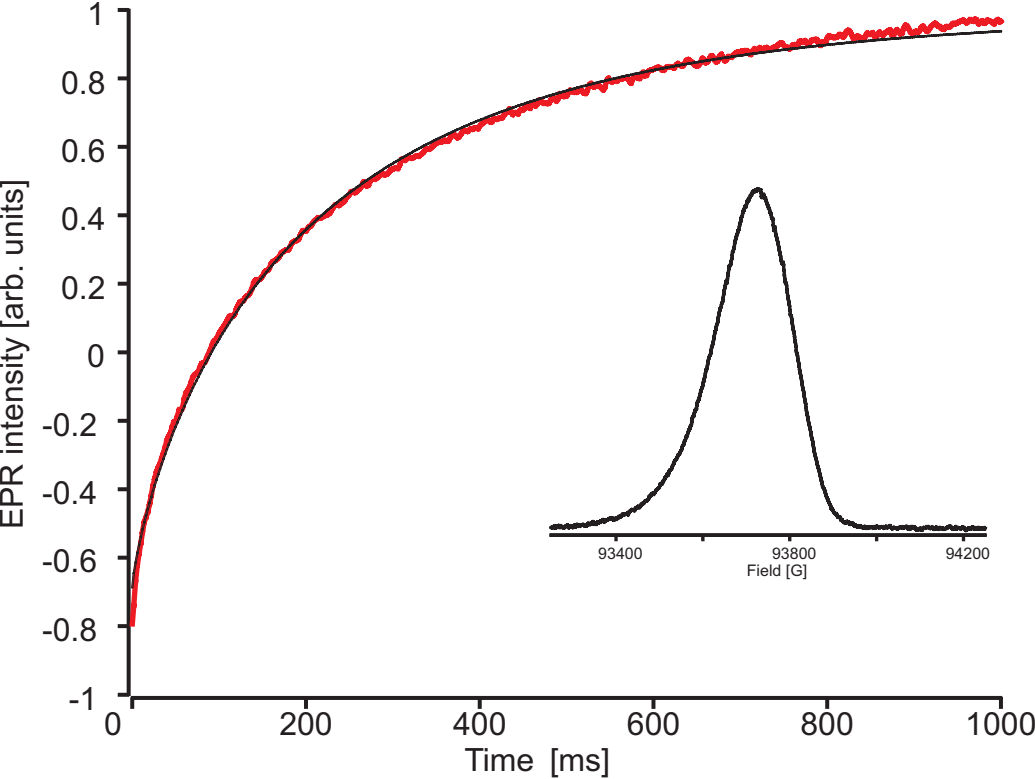

Microwave Bridge Tuning
MM-Band
Frequency:
Stand By
Tune
Operate
Bias:
Signal Phase:
Auto Tuning:
Up
Fine
Down
Stop
Reference Arm:
On
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Attenuation [dB]:
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Options
Log. Scale:
Monitoring
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Help

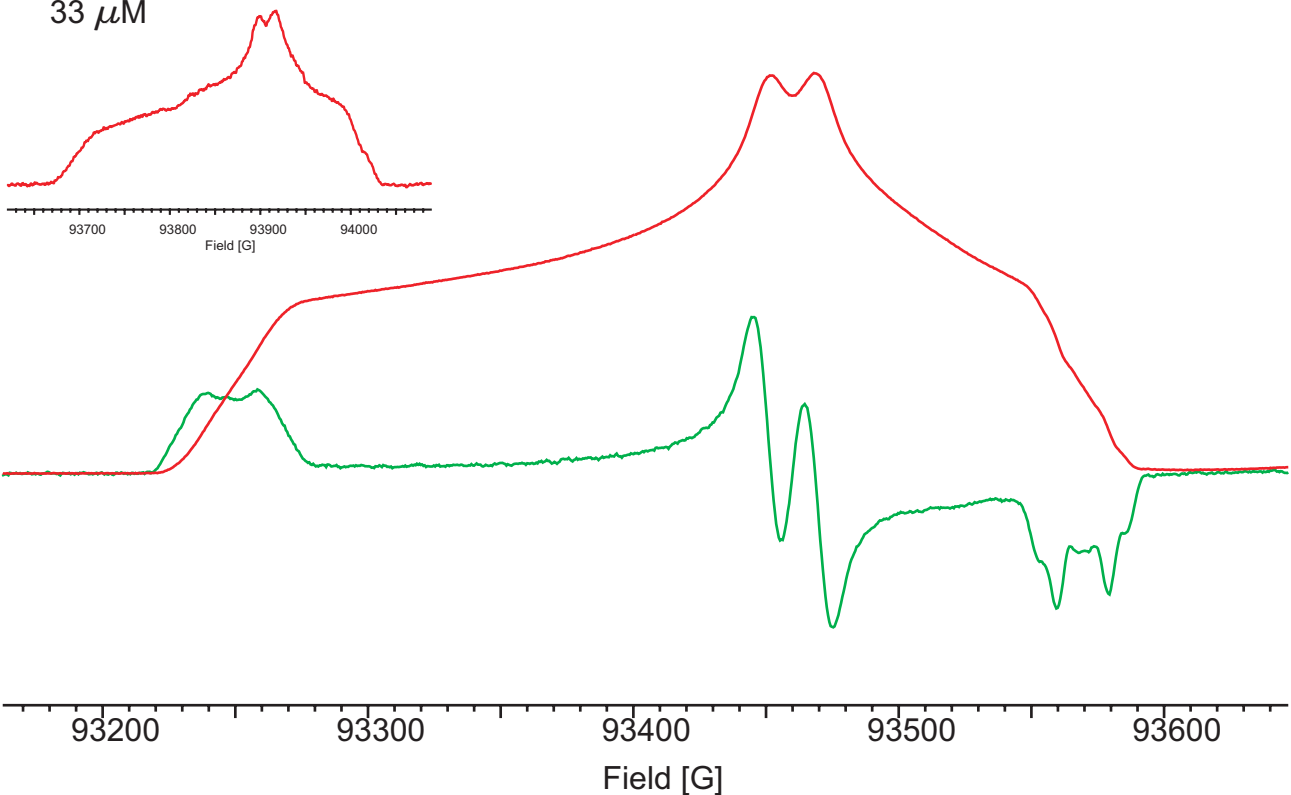

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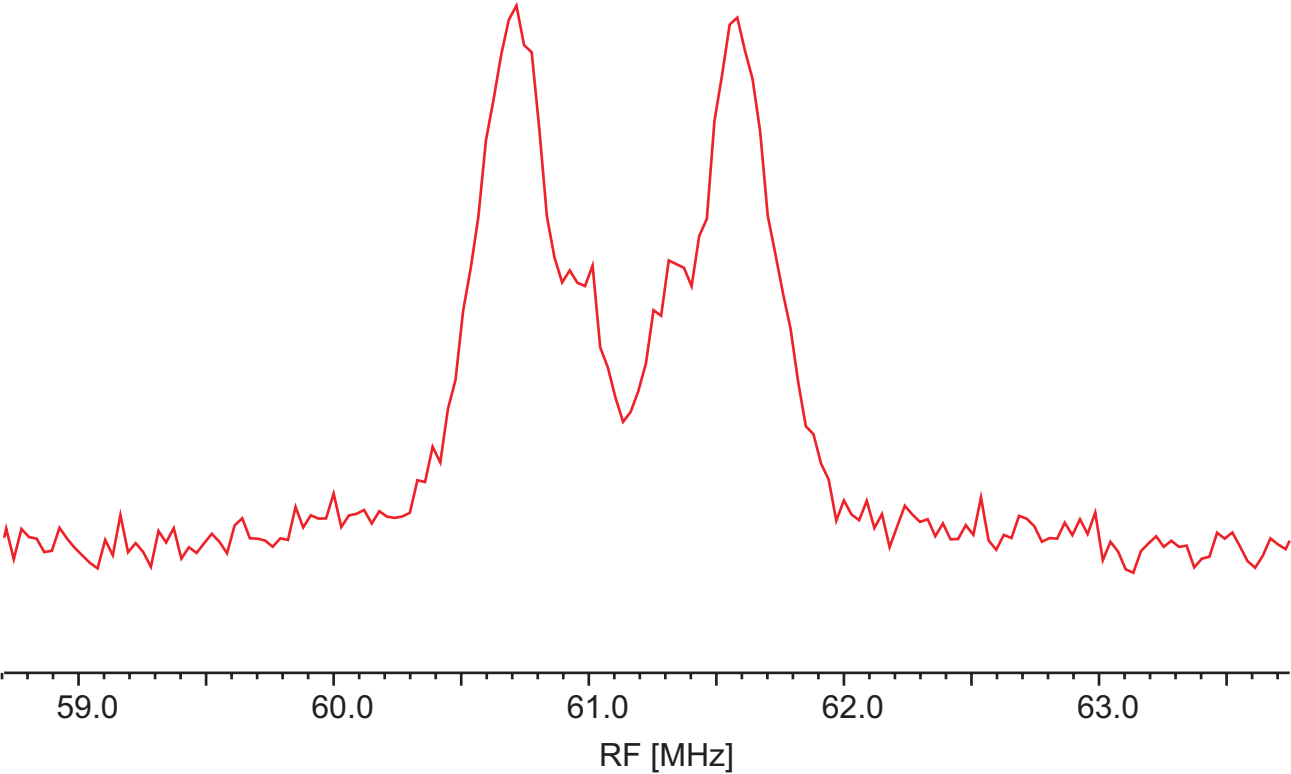

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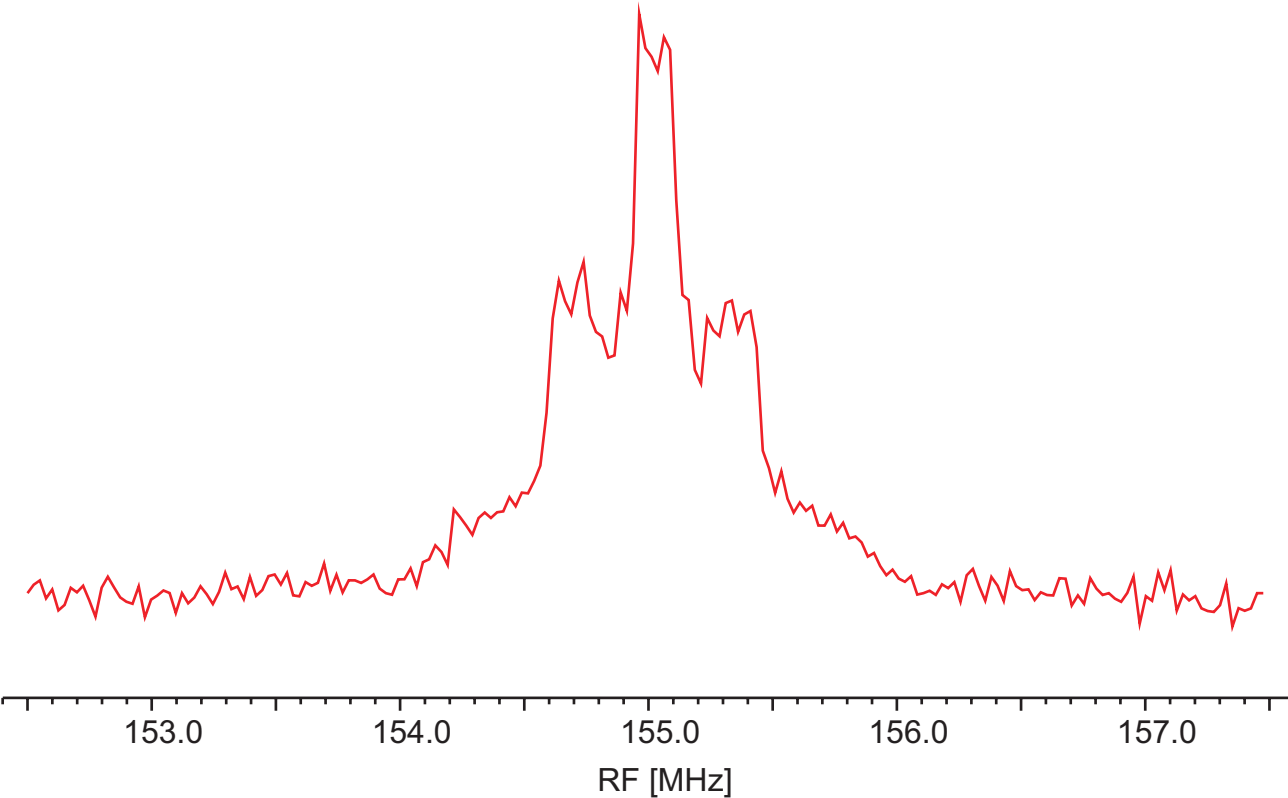

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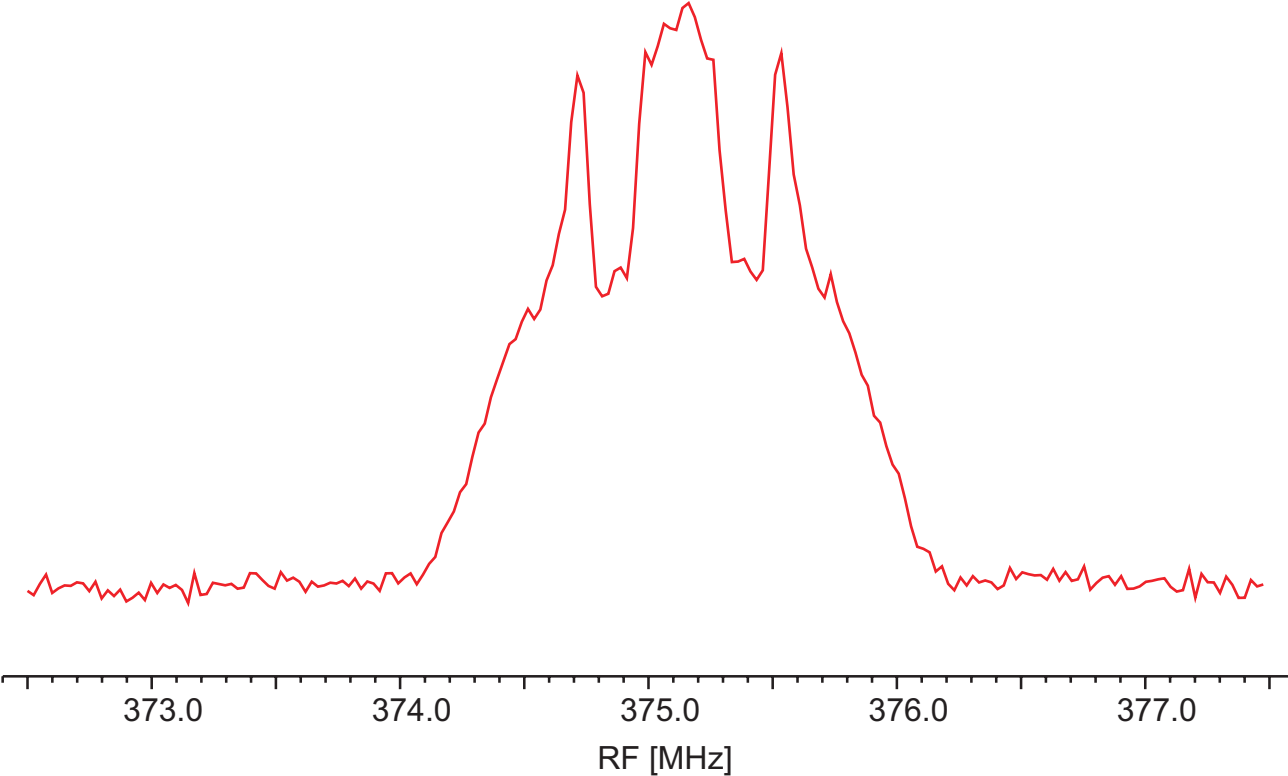

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1600
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2000
Time [ns]


EPR intensity [arb. units]
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0
-0.2
-0.4
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200
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Time [ms]
93400
93800
94200
Field [G]


33 μM
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93500
93600
Field [G]


59.0
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RF [MHz]


153.0
154.0
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RF [MHz]


373.0
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